From terminal to Claude Code

Let's set up a CLI AI agent do a task for you

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What we'll do in 45 minutes

1. **Learn agents in five minutes** — what's under the hood: a language model, a harness, and a loop
2. **Agents in the terminal** — Get a sense of what the command line interface (CLI) is, and how to start your own agent
3. **The exercise** — run a clinic audit, and look at how it's configured
4. **Final thoughts** — what you take away

> Everything you'll need to type or paste is in `copypaste.txt`, and a PDF of these slides is in the exercise folder — both inside the files we'll download in a bit.

Please interrupt me with questions / clarifications / if I talk too fast.

The chore we want to hand off

Every clinic, you check each patient against their **treatment target** - the goal of getting disease activity down to a low level.

- One patient at a time, by hand
- Tedious, and easy to slip on a busy day

> By the end of this session, that whole job is **one sentence** typed into a computer.
> And it will also have a dashboard to visualise the patients

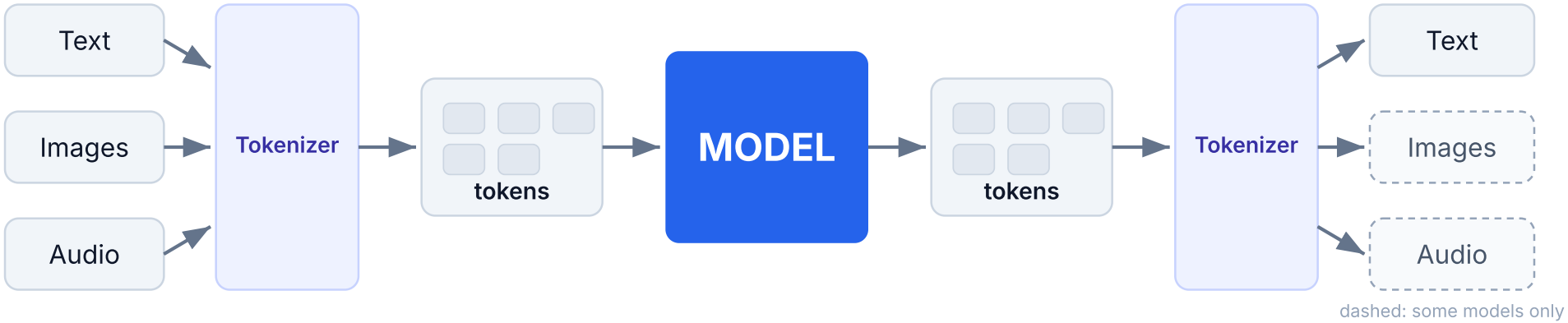
- Let's start with a 5-minutes overview of agents

What is an agent

- Think of an agent as someone who sits at your computer and can use it together with you.
- You can ask them to do things for you.
- Each agent has:
  - An LLM (brain)
  - A harness (body)
  - An agentic loop

The brain: a language model

A language model only predicts the next token.

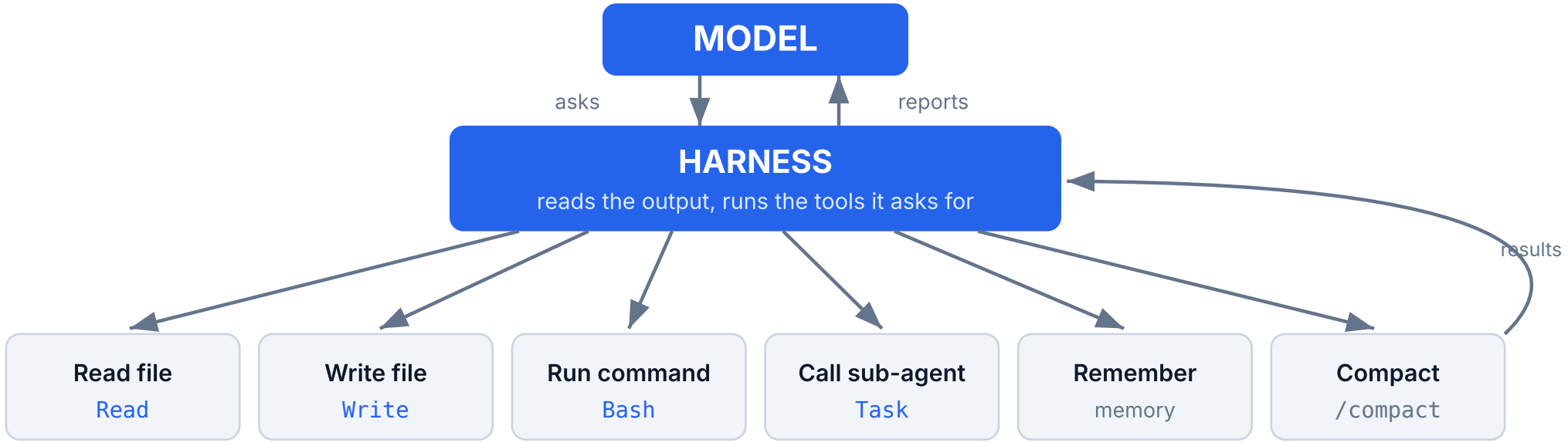


Brilliant at language, but only talk: no hands. A mind in a jar.

It can't reach your files: to read one, **you** paste it in; to *write* one, it tells you the text and **you** save it.

The body: a harness

The harness turns the model's words into actions.

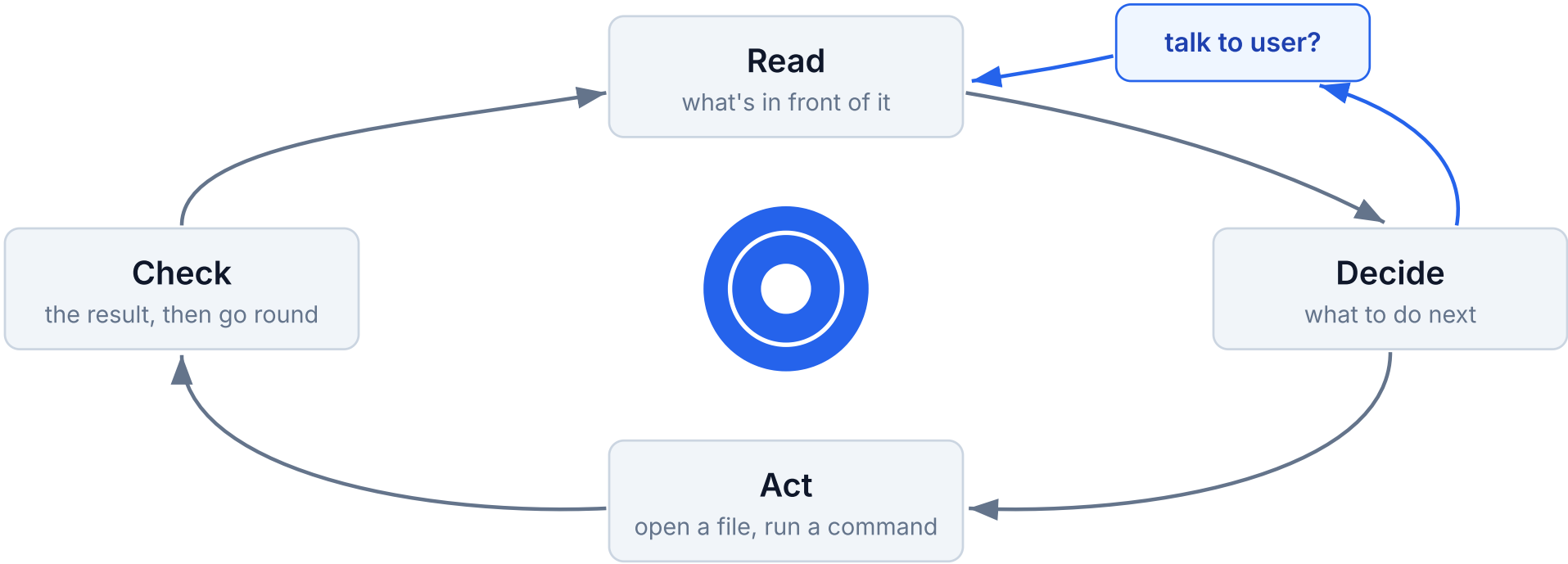


...and more (web search, and others) can be plugged in

Claude Code is a harness. So are **OpenAI Codex** and free open-source tools.
Cursor (next session) is a harness with buttons instead of typed commands.

The agent: a loop

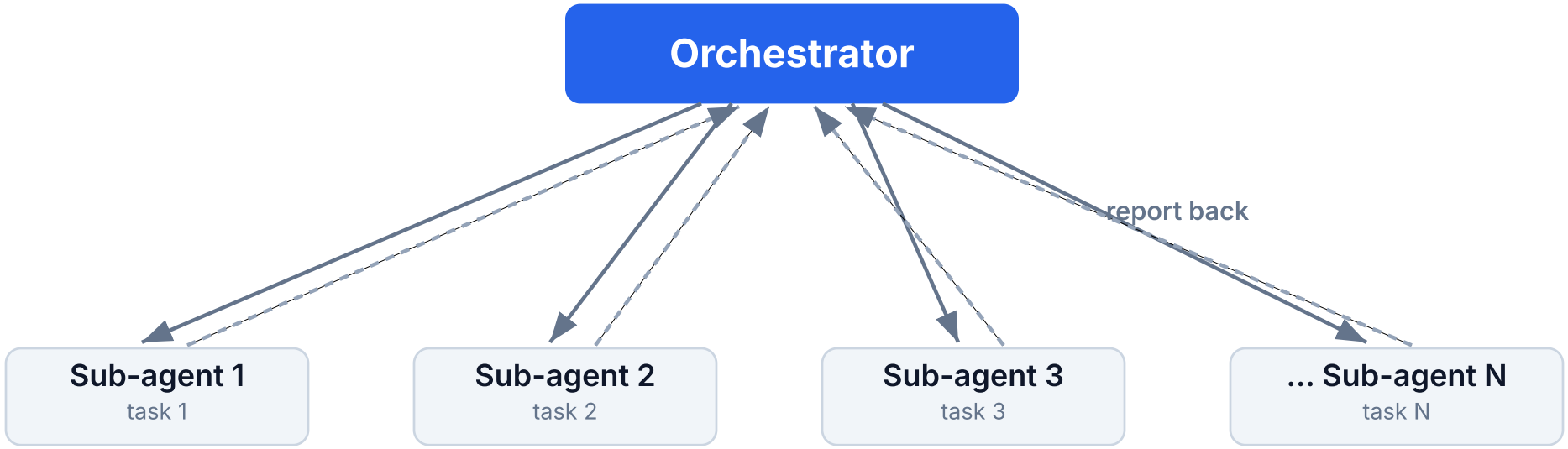
It keeps going round until the goal is met.



This loop, chasing a goal on its own, is an agent.

A swarm: many at once

One sub-agent per patient, all at once, reporting back.

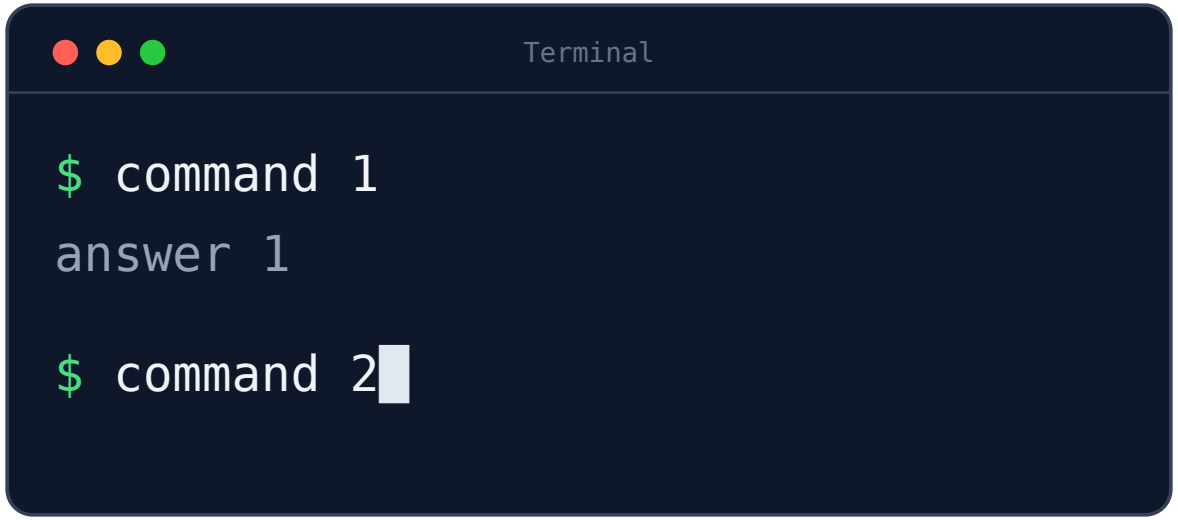


Each runs its own loop; you watch them side by side, like a board of jobs.

What is a terminal?

Before mice and windows, there was only *text* and keyboard. Old interaction mode: you type a command, it runs, the computer replies in text.

- A **direct, text way** to tell the computer what to do
- Everything you type is a command the computer executes
- One line in, one result out
- LLMs live inside text: most natural interface for them



Download the exercise

Get the patient letters and instructions from:

> `github.com/ganileni/clinic-audit-exercise`

- Open that page, click the green **Code** button, choose **Download ZIP**
- **Unzip** it somewhere easy to find

Inside you'll find `copypaste.txt` (every command and link, ready to paste) and a **PDF of these slides to follow along**.

Comfortable with git? Instead: `git clone https://github.com/ganileni/clinic-audit-exercise`

Open the terminal

- **Windows**: Start → type **PowerShell** → open
- **Mac**: Cmd+Space → type **Terminal** → open
- **Linux**: It depends (you should know if you're on Linux)

The few commands you need, the same in all three:

- `cd folder-name` — go into a folder
- `ls` — list what is in this folder
- `cd ..` — go back up one folder

A few more, harmless to try: `pwd` (which folder am I in?), `cat <file>` (show a file), `clear` (clean the screen), `date` (date and time).

> Now `cd` into the folder you downloaded — but **you don't need to type the path**, copy it: **Windows** right-click the folder → *Copy as path* → Ctrl+V; **Mac** select it → Cmd+Option+C → Cmd+V (or drag it into the window).

Set up and launch

Do this in two terminals — we launch two agents (the audit and the dashboard), so set each one up the same way.

You did these in the setup session.

- **Install Claude Code** (if you haven't): go to `code.claude.com` — it shows the install command for your operating system (with a link to the docs just below)
- `export ANTHROPIC_API_KEY=<the key on the slide>` — your access pass. **don't do this if you have your own account already set up**
- `export CLAUDE_CODE_SUBAGENT_MODEL=haiku` — per-patient readers run on cheap **Haiku**
- `claude` — opens **Opus** by default, which directs the work
- Press **Shift+Tab** to let it run on its own (**auto mode**)

NOTE: Every command on this slide is in `copypaste.txt`.

Send the prompt

Paste this and **hit enter**. The run starts now and works in the background:

> Follow the instructions in `AUDIT.md`. The patient letters are in `clinic/`. Write your output to `audit.json` in this directory.

That is the whole prompt. The rules, the output format, and the example all live in `AUDIT.md`.

It's better to write down long instructions in a file.

NOTE: Don't type it — this prompt is in `copypaste.txt`.

Send the dashboard prompt too

Now, in a new terminal, open a **second** agent and send this one, in parallel (remember to set it up and go into auto mode first):

> Follow the instructions in `DASHBOARD.md`. Build `dashboard.html`; it loads `audit.json` when the audit has written it.

Two jobs now run at once: the **audit** swarm, and the **dashboard** building alongside it.

This prompt is in `copypaste.txt` too.

They're running (about 8 minutes)

While the swarm audits and the dashboard builds, here is what we just asked

Why make a file called `AUDIT.md`?

The prompt is tiny. All the substance lives in one file, **`AUDIT.md`**.

- The prompt just says "follow `AUDIT.md`" (one sentence as I promised :))
- So the audit is **documented and reproducible**: anyone can read exactly what was asked
    - You can track changes and revert them (learn **version contol software** and **git**)
- You can **ask an LLM to draft** the document. You just correct the mistakes.
- This is the beginning of **a skill**, a repeatable workflow.

> Change the rule? Edit `AUDIT.md`, not a sentence you typed once and forgot.

Let's go through the document.

The document. The task and how we judge

The document begins with a description of the task, and then clearly introdces the definitions and the rule of the audit.

- **At target**: the disease activity score (DAS28-CRP) is **3.2 or below**
- A **breach** is all three together:
  - not at target, **and**
  - treatment was **not escalated**, **and**
  - **no reason** to hold off was written down
- Folic acid or a painkiller is **not** an escalation

Can't judge it cleanly? The agent flags **needs review**, instead of guessing

*A deliberately simplified teaching rule, not the full clinical standard. This needs to be iterated upon, and it's where **your expertise** comes in*

Prompting: elicited reasoning + grounding

A well-know technique to improve output

How it runs

For each patient, in order:

- **Gather** the records, which may be **split across several letters**
- One **sub-agent per patient**, all at once (the swarm)
- Keep the **exact quote and source** for every value

Watch for:

- Some letters give the raw numbers but **no score**
- The output is to **check, not trust**

What it delivers: a data file

The agent writes `audit.json`. *JSON* is just a simple, labelled data format that both people and computers can read.

- One entry per patient: the verdict, the reasoning, the evidence
- Every entry has the **same fixed shape** (a *schema*)

Why the *schema* fixed shape matters:

- Entries are **comparable** across patients
- A small **checker** can confirm nothing is missing or malformed
- Having the **schema** pre-specified, lets the dashboard agent know how the data looks like, so they can build a dashboard without ever having seen a piece of data.

Worked examples

Another proved method to improve output from an agent.

- Here we have 3,
- One is ambigous one to reiterate that the agent should not force a decision under ambiguity (they are usually tempted to)
- You can add examples for weird and corner cases where agents got things wrong in the past!

Show and tell: improving the prompt

You don't have to write every rule yourself. **Ask the agent to change the instructions for you.**

- Live: I ask it to add one rule to `AUDIT.md`: **build a checker and run it** at the end, so the audit verifies its own output
- It edits the instructions; the next run follows them

> Improve the recipe by talking to the agent, not just by typing.

It shows its sources

The audit has finished, and the dashboard we started earlier **fills with the results**: a **table of all patients** with their verdicts.

- Click a patient: the **reasoning**, and **every value with its quote and source letter**
- Click a source to open the **letter itself**
- A **review mode** steps through patients so you can **flag** the doubtful ones

> Fast, but you still trace it back and check. Do not blindly trust the output.

Where this leaves you

You handed off a clinic audit in one sentence, with every answer **sourced**.

- The prompt needs to be iterated. Find mistakes, update the prompt, use it again. You will build trust (and a sense of what to check) as you go.
- You are responsible for the final output, always check it or rigorously demonstrate that the system performs comparably with you.

Dive deeper:

- Prompting: `github.com/anthropics/prompt-eng-interactive-tutorial`
- Claude Code guide: `code.claude.com/docs/en/best-practices`
- Power-user tricks: `arps18.github.io/posts/claude-code-mastery`

All three links are in `copypaste.txt`, ready to open.

What we learned today

Today we met **agents**, the **terminal**, and **how to start an agent**.

Things to remember:

- Write your instructions down — that is a **skill** (a reusable recipe)
- A little **prompting** craft goes a long way
- Use an **agent to help you write the prompt**
- **Markdown** and **JSON**: plain text both you and the agent can read. Also check out YAML.
- Make it **easy to check the result**

What to do now:

- Review the slides and the materials when you have time
- Use LLMs to dive deeper in the concepts we discussed. You can ask them infinite questions!
- Learn by doing: try your hand at a different task.

THANK YOU

Questions?